

# PUNJAB RETROFITTING ROADS

**Sector:** Infrastructure  
**Hazard addressed:** Flood  
**Targeted assets:** Roads



**Description:** This measure involves retrofitting existing roads in flood-prone districts of Punjab to improve their resilience against flood hazards. Interventions include raising road embankments above the 1-in-100-year flood level, construction of retaining walls, and installation of additional culverts with enhanced flow capacity to allow floodwater to pass without road damage.

**Priority Districts:** Mianwali, Chakwal, Gujrat, Gujranwala, Bahawalpur, Rahim Yar Khan, Lahore

## TECHNICAL SPECIFICATIONS/DIMENSION

**Unit of implementation:** Per km of road.

**Scale assumptions:** Roads located in flood prone areas of the target districts.

**Key design parameters:** Raising road embankments above 1-in-100-year flood level; construction of retaining walls; installation of additional culverts with enhanced flow capacity.

**Time horizon:** 2026–2035

## IMPACT

**Risk reduction level:** Very High Effectiveness – large reduction in damages (Expert Rank 1).

**Time to effectiveness:** Effectiveness begins progressively as individual road sections are retrofitted over 2026–2035, with full effectiveness from 2036 onwards.

**Expected impact:** 90% reduction in road damage (MDD = 0.1); protects road asset base valued at USD 418,727,992 across seven target districts.

## IMPLEMENTATION CONSIDERATIONS

**Enabling conditions:** Road asset inventory and condition survey; hydraulic design for embankment raising; procurement of qualified and experienced contractors; traffic management plans during retrofits/rehabilitation.

**Required institutions:** Punjab Communications & Works (C&W) Department; National Highway Authority (NHA); District governments and Traffic Police.

**Barriers:** Procurement complexity for large-scale road works; traffic disruption during construction; coordination gaps between Punjab C&W, NHA and district governments/police; budget release timing; access to flood-prone road sections during construction.

## COST ASSUMPTIONS

**Unit cost:** USD 13,278 per km.

**Capital Expenditure (CAPEX):**  
USD 83,745,598.

**Operational Expenditure (OPEX):**  
USD 133 per km per year.

## CO-BENEFITS & TRADE-OFFS

**Environmental:** Properly identified and designed culverts allow natural water flow across roads without damming.

**Social:** Maintained road connectivity during floods enables emergency response and evacuation; continued market access for rural communities; reduced isolation of flood-affected areas.

**Economic:** Avoided road reconstruction costs damaged due to floods; maintained supply chains and freight movement; reduced indirect economic losses from transport disruption.

**Potential risks:** Traffic disruption during retrofitting works; access constraints in seasonally flooded areas; residual risk for events exceeding 1-in-100-year threshold; road maintenance quality affects long-term effectiveness.

## BENEFICIARIES

**Primary beneficiaries:** Communities dependent on road connectivity for livelihoods, access to education and health facilities; emergency response agencies.

**Distributional aspects:** Particularly beneficial for rural and agricultural communities in flood-prone areas of target districts with limited alternative transport routes.